Technical Document #322: Cloud Computing - Comprehensive Overview

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Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Microservices architecture structures applications as a collection of loosely coupled services.
- Multi-cloud strategies use services from multiple cloud providers.
- Platform as a Service (PaaS) provides a platform for developing and deploying applications.